

Project 1

Name of the game: Tank War

Author:

Lim Yik Seng (E24105038), Meng Ting Hsu (E24106830),
Kor Yu Ee (E24105062), Zhang Wei Lun(E24106157)

Date: May, 10, 2022

Compilation guide:

1. Open terminal 1 and type enter "cd" for entering the file.
2. Enter "qmake-qt5 -project". (Generate .pro file)
3. Enter "qmake-qt5 "QT+=widgets" "QT+=core gui" "QT+=widgets multimedia"". (Generate Makefile)
4. Enter "make".
5. Open terminal 2 and enter "startxwin".
6. execute /bin/pulseaudio.exe in Cygwin download folder by administrator.
7. Open terminal 1 and enter "export DISPLAY=:0.0".
8. Enter "./mid" for execution.

Target of this game: In order to protect your Eagles(Base Camp) from getting destroyed by the enemy tank. Therefore, you need to control your tank to destroy all of the enemy tank.

- **Our tank** -> The tank that the player can control by himself/herself. Once our tank is destroyed, the game is over.
- **Enemy tank** -> The tank that will go ahead to destroy the eagles.
- **Stronger enemy(red tank)** -> The tank that its health point is more than the normal enemy tank.
- **Bullet** -> The player need to control the player's tank to shoot the bullet in order to destroy those enemy tanks.
- **Eagles(Base Camp)** -> The base that the player needs to protect from being destroyed by the enemy tank. Once the base camp is destroyed by the enemy tanks, the game is over.
- **Health point(on the scoreboard)** -> Our tank : 300 Hp, Base camp: 100 Hp, Enemy tank: 100 Hp, Stronger enemy(red tank): 300 Hp.
- **Number of enemies left(on the scoreboard)** -> The enemies left in this stage.
- **Number of stage(on the scoreboard)** -> The current level of stage.
- **Design background** -> Bricks, iron bricks, grass, sea, background colour.

Controls:

➤ **Start the game (at every stage) ***Need to press the button to start the game or else the game will not start**

stage 1,4 -> Number pad 1

stage 2,5 -> Number pad 2

stage 3,6 -> Number pad 3

➤ **Stop the game**

stage 1~6 -> Number pad 0

➤ **Movements:**

1P -> up: W 2P -> up: Number pad 8

-> down: S -> down: Number pad 5

-> left: A -> left: Number pad 4

-> right: D -> right: Number pad 6

➤ **Shoot bullets:**

1P-> F 2P-> Number pad 7

➤ **Background Sound Effect:**

Start the background sound effect -> O

Stop the background sound effect -> M

Bufs:

➤ **Frozen buff:**

-> It will freeze all of the enemy tanks in the map for 10 seconds, but the enemy tanks are still able to shoot the bullets, be careful!

-> When you take the frozen buff during the time duration of the previous frozen buff, the ability of the buff will not be generated.

➤ **Weapon upgrade buff:**

a. When getting the buff for the first time, the bullets speed double.

b. When getting the buff for the second time, our tanks can shoot two bullets at the same time.

c. When getting the buff for the third time, the bullet can break the iron bricks.

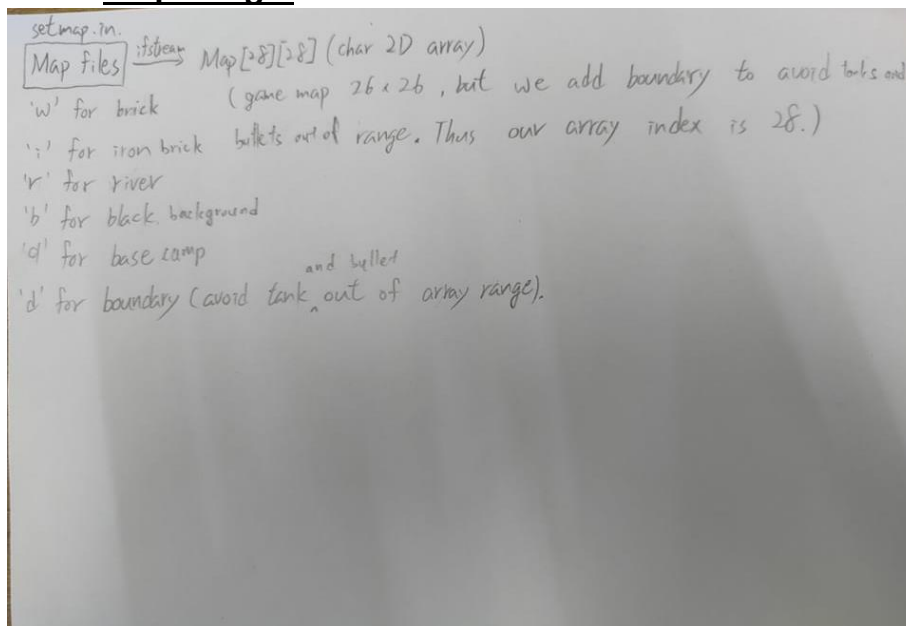
➤ **Explode buff:**

-> When you have taken the buff, all of the enemy tanks on the map will be exploded.

-In this game, the player needs to control the player's tank to destroy the enemy tanks in order to protect the eagles from being destroyed by them. If the enemy tanks are destroyed completely and the eagles haven't be destroyed, then you win the game. Otherwise, either the eagles or your tank is destroyed by the enemy tank, you lose the game.

Implement Specification in Detail:

1. Map design:

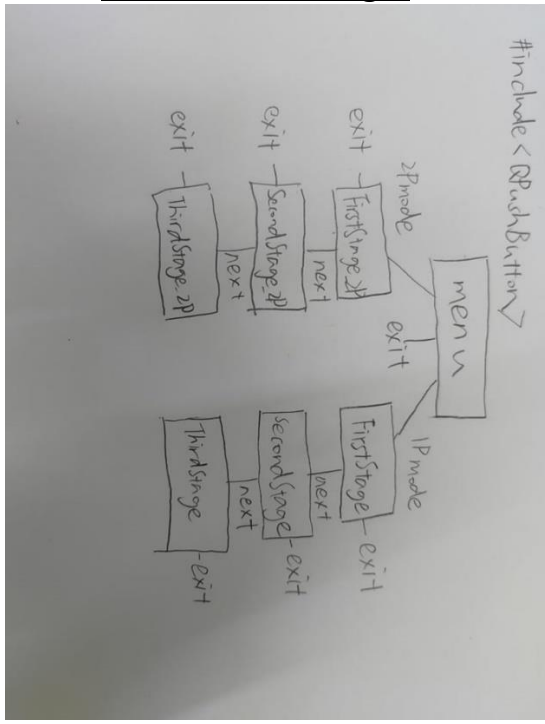


- We have designed an 2D array Map[28][28], the actual game map is actually 26*26 just like the basic requirements that TAs gave, but we also have designed a boundary(invisible, it has the same colour as the interface background) to surround the game map for the

convenience of coding in the part of tank movement and bullet shooting to prevent tanks and bullets go beyond the game map.

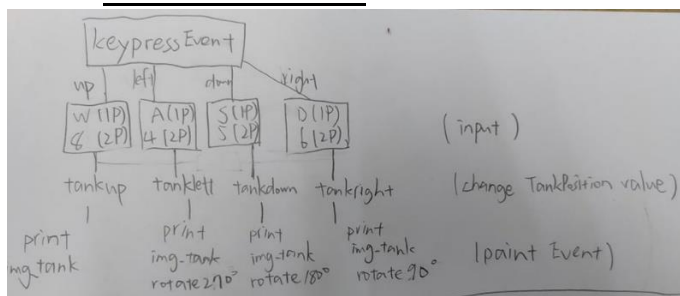
- In our map design, 'w' is brick, 'i' is iron brick, 'r' is sea, 'b' is black background, 'q' is base, 'd' is boundary(white background, same colour as the interface background).
- The compiler will read our map via ifstream code, and every stages have its own map file simply for the fact that the design of every stages are different.

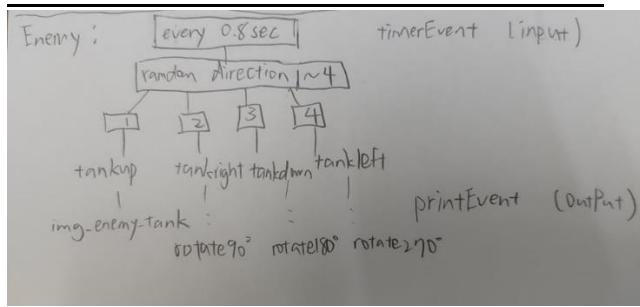
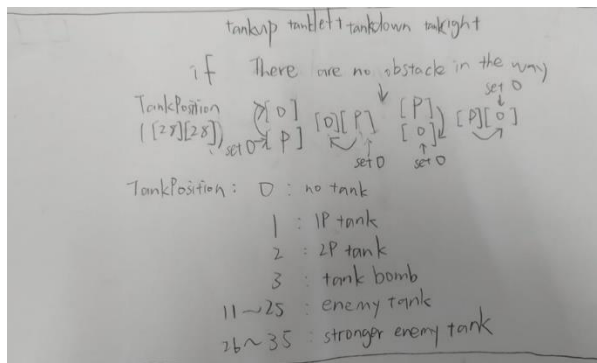
2. Game menu design:



- `#include <QPushButton>` is needed for this part.
- We have designed a game menu for you to chose the mode of game(1P/2P) at first and also have the exit button for exit the game.
- Every stages have a button for exit and a button "next" for going to the next stage.

3. Tank movement:





➤ Our tank:

- In keypressEvent, we have designed the code for the movement of our tank(including 2P), when the tank moves, it will change the tank position(TankPosition) of our tank at first and the paintEvent will paint a new image of tank at the place that tank has moved to and the image of the tank before moved will be gone simply for the reason that the tank position is already changed.
- ❖ When the tank moves upwards, the image that will be painted is the normal tank.
- ❖ When the tank moves downwards, the image that will be painted is the tank image that already be rotated 180°.
- ❖ When the tank moves to the right, the image that will be painted is the tank image that already be rotated 90°.
- ❖ When the tank moves to the left, the image that will be painted is the tank image that already be rotated 270°.

***The definition of our rotation is that rotate clockwise.

➤ Enemy tank:

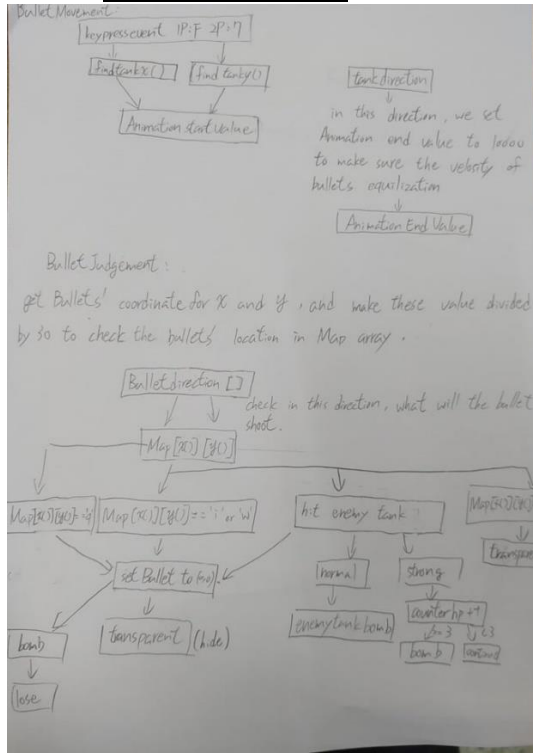
- The code that we designed for the enemy tanks is that the enemy tanks will move automatically every 0.8 seconds(in timerEvent), it will have 20% to change the direction(1, 2, 3, 4)(up, right, down, left) and move one step(30*30 pixels). The way of paint the image of enemy tanks is same as "our tank".

➤ Remark:

- ✓ We have designed an array TankPostion[35] for all of the tanks(including our tank, enemy tanks)
 - ❖ TankPostion[0]: no tank
 - ❖ TankPosition[1]: 1P tank
 - ❖ TankPosition[2]: 2P tank
 - ❖ TankPosition[3]: tank boom(image of the tank boom)
 - ❖ TankPosition[4-10]: no tank
 - ❖ TankPosition[11-25]: enemy tanks
 - ❖ TankPosition[26-35]: stronger enemy tanks
- ✓ Tank movement checking (inside tankup, tankleft, tankdown, tankdown function): If there is no obstacle(sea, bricks, iron bricks) in the path that the tanks(all of the tanks) move, and then the tanks can move, or else it will be blocked.

- ❖ The code that we have designed is if there is obstacle in front of the path, the code of tank movement(TankPosition) will not be executed and if there is no obstacle, the tank will move, the code and the idea is already written above in “our tank” and “enemy tank”.

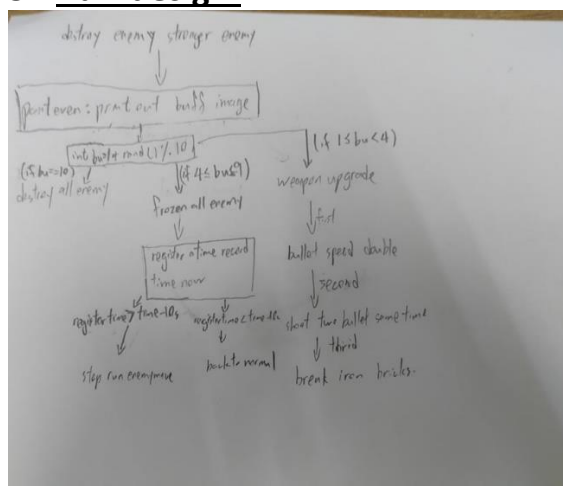
4. Bullet shooting:



- We let the bullet be transparent when we load the image in constructor, it is because if we did not, the bullet image will appear at the upper left corner even if we did not set the start value yet.
- In keyPressEvent, we have designed the code for the bullet to be shot. The key for shooting the bullet is 'F'(1P), number pad 7(2P).
- ❖ We have designed two function(findtankx(),findtanky()) in order to find the coordinate of tank for the starting point of the bullet and the bullet movement is designed with animation (animation duration, animation start value, animation end value, animation start()).
- ❖ The animation end value, we set a extremely far coordinate(10000 or -10000) and the duration we set is the same, this is to ensure the bullet speed is almost the same. If we did not set the end value and duration like this way, if the tank is near to the left and shoot the bullet to the left the bullet speed will extremely slow (if the end value is set to the coordinate of the end point to the left of the game map) and if the tank is far from the right and shoot the bullet to the right, the bullet speed will obviously be faster than the bullet speed to the left (if the end value is set to the end point to the right of the game map).
- Bullet movement checking :
- ❖ Using the bullet->x(), bullet->y() function to get the coordinate of the moving bullets(all of the bullets on the game map) and these values are divided by 30 (because 1 grid = 30*30 pixels) to check the bullets' location in our Map array.
- ❖ We have designed an array (Bulletdirection[50]) (the size of array is declared to 50 because every tanks in the map have its own bullet, and every bullets have its own bullet direction).
- ❖ (in bulletJudgement function)

- ✓ If the next step of the bullets is base camp(Hp=100), then the game is over.
- ✓ If the next step of the bullets is bricks or iron bricks or enemy tanks or stronger enemies or boundary, we set the bullet image to the coordinate (0,0) and hide the image(in bulletstop function), if we do not set the bullet image to another place and let the picture hide only and stay at the original place, when the enemy tanks pass through that place our "bullet checking" will assume that the bullet hits the enemy tanks and the enemy tanks will destroy.
- ✓ Our tank's bullet
 - If the next step of the bullets is enemy tank, then the enemy tank will be destroyed and the bullet will vanish and the sound effect of tank boom will be generated.
 - If the next step of the bullets is stronger enemies(Hp=300), and if the stronger enemies have not be hit for three times, the bullet will vanish, and the stronger enemies still survive, and if the stronger enemies have already be hit for three times the bullet will vanish and the tank will be destroyed and the sound effect of tank boom will be generated. The hp of stronger enemies is recorded in int array counter_hp[26-35].
- ✓ Enemy tank's bullet
 - If the bullets are from the enemy tanks and the next step of its bullets is our tank, then our tank will be destroyed and the bullet will vanish and so the sound effect of the tank boom and game over will be generated.
- ✓ If the next step of the bullets is bricks, the bricks will be destroyed and the bullet will vanish.
- ✓ If the next step of the bullets is iron bricks or boundary, the bullet will vanish.
- ✓ If the next step of the bullets is sea, the bullet will directly pass through.
- ✓ If the next step of the bullets is grass, the image of the bullet will be transparent(in bullethide function) and yet the bullet is still moving.
- bulletload function is for the loading of the bullet image for all of the tanks in the game map.

5. Buff design:



- In timerevent, if it detect stronger enemy bomb(tankposition=4);it will randomly spawn 3 buff :frozen (buffposition=1), weapon upgrade(buffposition=2) ,explosion(buffposition=3).
- Use buffcheck function, if ourtank's position is equal to buff's position, we will get buff.

- When all our tanks/our tank or our base camp are destroyed, it will play a “lost” sound effect.

Troubles of during programming:

- At first, we are not familiar the qt5, so it has caused a lot of troubles and our time is being wasted so much.
- There is no error when compiling, but the game cannot be opened at all (core dumped) and at last we found that it is because it has exceeded the size of our array we declared.
- We could not let the bullet be invisible(transparent).
- We could not let the bullet vanishes when the bullet already shoots one target and so the bullet can shoot all the targets in that line.
- We were hesitating about the movement of the bullet and the tank, in using animation or painter.